

Module 7: Molecular Genetics — Study Questions

1. What are the three parts of a nucleotide?
2. What are the base pairing rules for DNA? (A pairs with , *G pairs with*)
3. What does "semi-conservative" mean in DNA replication?
4. What does helicase do during DNA replication?
5. What does DNA polymerase do during DNA replication?
6. What is the difference between the leading strand and the lagging strand?
7. If a DNA strand reads 5'-ATCG-3', what is the complementary strand?
8. What is transcription? Where does it happen?
9. How is RNA different from DNA? (Name two differences.)
10. Given the DNA template strand 3'-TACGGC-5', write the mRNA.
11. What are the three types of RNA and what does each do?
12. What is a codon? How many bases does it have?
13. What is the start codon? What amino acid does it code for?
14. Name the three stop codons.
15. What is the job of tRNA during translation?
16. What happens at the A site, P site, and E site of the ribosome?
17. Using a codon table, translate this mRNA: AUG-GCA-UAA. What is the amino acid sequence?
18. What is a point mutation? What is a frameshift mutation?

19. Why are frameshift mutations usually more harmful than point mutations?

20. Summarize the Central Dogma: DNA $\rightarrow$ $\rightarrow$